Rooted-tree summation and Catalan closure for polymer cluster expansions with holes: a Lean 4 formalization blueprint

Lluís Eriksson

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Abstract

We isolate the combinatorial engine that converts a first polymer activity $K^{\#}$ into the second Ursell activity $H^{\#}$ in a polymer expansion with holes. The fixed-union condition, a marked active root, hard-core incompatibility, and the hole-modified metric are retained until the target decay has been extracted. The local leaf summation for a rooted labelled tree T produces the exact factor $M^{2n+1}\prod_v c_T(v)!$, where $c_T(v)$ is the number of children of v in the root-0 orientation. We prove the exact identity

$$\frac{1}{n!}\sum_{T\in\mathcal{T}_{n+1}}\prod_{v=0}^{n} c_T(v)! = \mathrm{Cat}_n$$

by ordering the children at every vertex and identifying the resulting objects with labelled plane rooted trees. This replaces a geometric closure by the Catalan majorant

$$B_M(\varepsilon)=\sum_{n\geq 0}\mathrm{Cat}_n\, M^{2n+1}\varepsilon^{n+1}=\frac{1-\sqrt{1-4M^2\varepsilon}}{2M},\qquad B_M = M\varepsilon + MB_M^2.$$

For $0\leq 4M^2\varepsilon<1$, this bound is coefficientwise sharper than $G_M(\varepsilon)=M\varepsilon/(1-4M^2\varepsilon)$. We transport the estimate through a rooted modified-metric profile and into the scalar one-scale remainder predicate `SingleScaleUVDecay`. The result specifies an Appendix-F-style cluster-expansion mechanism and the corresponding Lean 4 theorem manifest; it does not construct the Yang–Mills activity, a continuum limit, Osterwalder–Schrader reconstruction, or a Yang–Mills mass gap.

Keywords. polymer cluster expansion; Ursell functions; rooted trees; Catalan numbers; polymers with holes; renormalization group; Lean 4; formalization blueprint.

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Contents

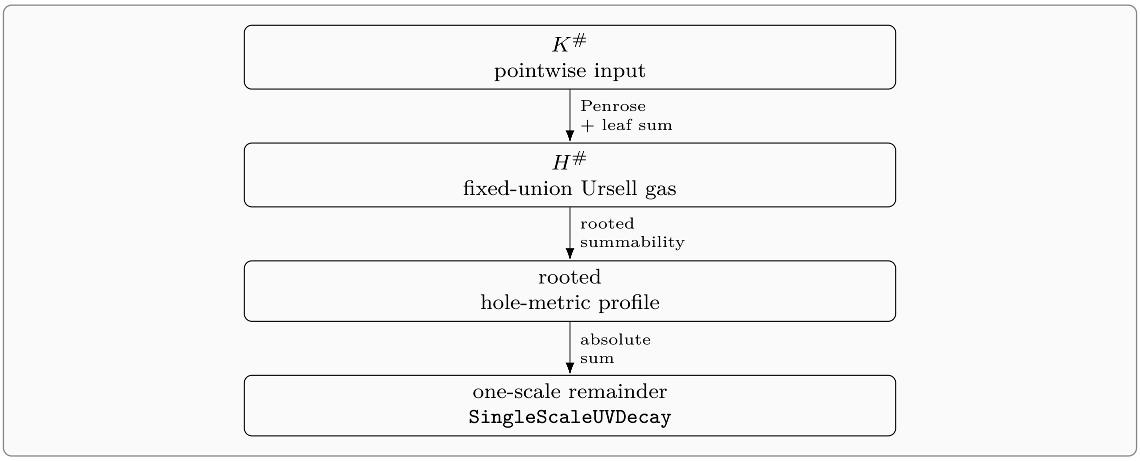
1 Introduction

Cluster expansions turn a partition function into an exponential of connected activities in abstract polymer systems [7]. In the presence of holes, however, the bookkeeping has two support maps: a full target support, whose union must remain prescribed, and an active skeleton, which controls incompatibility and rooted summability. Appendix F of Dimock's presentation of the Balaban renormalization group is a canonical source for this structure [2]. Its conclusion has the schematic form

$$|H^{\#}(Y)| \le CH_0 \exp[-(\kappa - 3\kappa_0 - 3)d_M(Y \bmod \Omega^c)],$$

provided a small first activity is available and the hole-modified geometry is summable. The purpose of the present paper is not to reprove the model-specific production of that first activity. It is to isolate, sharpen, and state a machine-checkable interface for the second-gas combinatorics that starts once $K^{\#}$ has been supplied.

The referenced Lean development already contains a two-support target compiler, an integrated first activity, the second Ursell object, Penrose tree domination, a fixed-tree leaf recursion, a geometric 4^n closure, rooted hole-metric summability, and a consumer-facing one-scale remainder bridge [4]. Its fixed-tree estimate is sharper than its final tree summation: for a labelled tree with $n+1$ vertices it retains $\prod_v c_T(v)!$, but the normalized sum over tree shapes is subsequently bounded by 4^n. The main observation of this manuscript is that this last step is exactly Catalan.



The contribution is a closure theorem rather than a new discovery about Catalan numbers. The identity is classical; what matters is its exact placement inside a fixed-union expansion with holes and its transport through a reusable formal API. The paper proves four linked statements.

1. Local hard-core moment estimates imply, for every fixed rooted labelled tree, the factor $M^{2n+1}\prod_v c_T(v)!$ while preserving the marked root and the parent-edge incompatibility indicators.

2. Ordering each child fiber converts the weighted tree sum exactly into the number of plane rooted trees with labelled nonroot vertices. The normalized sum is Cat_n, not merely bounded by 4^n.

3. The complete second-Ursell series is bounded by the square-root majorant B_M, the least nonnegative solution of $B = M\varepsilon + MB^2$.

4. The pointwise target estimate implies a rooted profile and, under the existing summability hypothesis, the scalar one-scale remainder estimate used by the one-scale consumer predicate.

The formalization boundary is organized so that the pure tree theorem can be reused outside the Balaban setting. The tree enumeration does not import gauge geometry, while the real square-root closure does not import polymers. This separation is important for both mathematical clarity and artifact review.

2 Polymers with holes and the second gas

We state the finite-volume theorem in an abstract form that matches the Lean interfaces. All sums in the combinatorial part are finite. Infinite-volume or scale-uniform statements enter only through explicitly stated summability hypotheses.

2.1 Two supports and a modified metric

Let Λ be a finite block set, let $\Omega \subseteq \Lambda$ be the active region, and write Ω^c for the holes. A polymer $Q \in \mathcal{P}$ carries

- a full support $\operatorname{supp}(Q) \subseteq \Lambda$;
- a nonempty active skeleton $\operatorname{skel}(Q) \subseteq \Omega$;
- a shifted hole-modified size $\ell(Q) = d_M(Q \bmod \Omega^c) + 1 \in \{1, 2, \ldots\}$.

The hard-core incompatibility relation is

$$Q \not\sim Q' \quad \Longleftrightarrow \quad \operatorname{skel}(Q) \cap \operatorname{skel}(Q') \neq \varnothing.$$

For a tuple $\boldsymbol{Q} = (Q_0, \ldots, Q_n)$, its prescribed full target is

$$U(\boldsymbol{Q}) = \bigcup_{i=0}^{n} \operatorname{supp}(Q_i).$$

The skeleton of a representable target Y is assumed to satisfy

$$\operatorname{skel}(Y) = \bigcup_{i=0}^{n} \operatorname{skel}(Q_i) \qquad \text{whenever } U(\boldsymbol{Q}) = Y.$$

Thus admissibility and incompatibility use active skeletons, whereas the fiber of the second gas uses the full union Y.

Assumption 2.1 (Metric stitching)**.** If the incompatibility graph of $\boldsymbol{Q}$ is connected and $U(\boldsymbol{Q}) = Y$, then

$$\ell(Y) \leq \sum_{i=0}^{n} \ell(Q_i). \tag{S}$$

In unshifted notation, (S) has the familiar shape $d_M(Y) \leq \sum_i d_M(Q_i) + n$. The shift by one is useful because it makes the multiplicative decay extraction compatible with the number of tree edges.

2.2 First activity and local moment budgets

Fix a residual rate $\rho \geq 0$, a leaf rate $\kappa_0 > 0$, a small parameter $\varepsilon \geq 0$, and a nonnegative weight $u : \mathcal{P} \to [0, \infty)$. Assume the first activity obeys

$$|K^{\#}(Q)| \leq \varepsilon\, e^{-\rho \ell(Q)} u(Q). \tag{1}$$

In the concrete hole geometry, $u(Q)$ is dominated by a spare exponential such as $e^{-2\kappa_0 \ell(Q)}$. We isolate only the two consequences consumed by the tree recursion.

Assumption 2.2 (Child and root moment budgets)**.** There is $M > 0$ such that, for every $j \in \mathbb{N}$, every parent polymer Q, and every active block $r \in \Omega$,

$$\sum_{Q' \not\sim Q} \frac{\ell(Q')^j u(Q')}{\ell(Q)} \leq j!\, M^{j+1}, \tag{C$_j$}$$

$$\sum_{Q:\, r \in \operatorname{skel}(Q)} \ell(Q)^j u(Q) \leq j!\, M^{j+1}. \tag{R$_j$}$$

The factorial dependence is the decisive feature: the moment index will become the number of children of a tree vertex. The repository's concrete constant is named `append ixFSecondUrsellMomentConstant`.

2.3 Ursell coefficient and exact-union activity

Let

$$f(Q, Q') = -\mathbf{1}[Q \not\sim Q'].$$

For $m \geq 1$, the Ursell coefficient is

$$\varphi^T(Q_1, \dots, Q_m) = \sum_{G \in \mathcal{C}_m} \prod_{\{i,j\} \in E(G)} f(Q_i, Q_j), \tag{2}$$

where $\mathcal{C}_m$ is the set of connected simple graphs on $\{1, \dots, m\}$. The second-gas activity with prescribed full union is

$$H^\#(Y) = \sum_{m \geq 1} \frac{1}{m!} \sum_{\substack{(Q_1, \dots, Q_m) \in \mathcal{P}^m \\ \bigcup_i \operatorname{supp}(Q_i) = Y}} \varphi^T(Q_1, \dots, Q_m) \prod_{i=1}^m K^\#(Q_i). \tag{3}$$

Write $m = n + 1$ and denote the corresponding order by $H_n^\#(Y)$.

The hard-core tree-graph inequality gives

$$|\varphi^T(Q_0, \dots, Q_n)| \leq \sum_{T \in \mathcal{T}_{n+1}} \prod_{\{i,j\} \in E(T)} \mathbf{1}[Q_i \not\sim Q_j], \tag{4}$$

where $\mathcal{T}_{n+1}$ is the set of labelled spanning trees on $\{0, 1, \dots, n\}$. We use only this inequality; sharper Penrose partition schemes remain compatible with the argument [1, 5].

3 Preserving the target and marking a root

The order of the estimates is essential. We first retain the exact-union condition, use it to extract target decay, and only then enlarge the fiber to a root-marked sum over all tuples.

Lemma 3.1 (Target decay before overcounting)**.** *Let $T \in \mathcal{T}_{n+1}$ and suppose all its parent-edge incompatibility indicators are nonzero. If $U(\boldsymbol{Q}) = Y$, then*

$$\prod_{i=0}^n e^{-\rho\ell(Q_i)} u(Q_i) \leq e^{-\rho\ell(Y)} \prod_{i=0}^n u(Q_i).$$

Proof. The nonzero tree indicators make the incompatibility graph connected. Assumption 2.1 gives $\ell(Y) \leq \sum_i \ell(Q_i)$. Since $\rho \geq 0$, $e^{-\rho \sum_i \ell(Q_i)} \leq e^{-\rho\ell(Y)}$. □

Choose $r \in \operatorname{skel}(Y)$. Every tuple in the exact-union fiber satisfies

$$1 \leq \sum_{i=0}^n \mathbf{1}[r \in \operatorname{skel}(Q_i)]. \tag{5}$$

The summand in (4) is invariant under simultaneous relabelling of the tuple and tree. All $n + 1$ marked coordinates therefore contribute equally, and

$$\frac{n+1}{(n+1)!} = \frac{1}{n!}.$$

After this step the marked coordinate may be placed at label 0, and the exact-union restriction may be dropped because all remaining terms are nonnegative. The target factor $e^{-\rho\ell(Y)}$ has already been extracted.

Proposition 3.2 (Rooted reduction of a fixed order)**.** *For $r \in \mathrm{skel}(Y)$,*

$$e^{\rho\ell(Y)}|H_n^{\#}(Y)| \le \frac{\varepsilon^{n+1}}{n!} \sum_{T\in\mathcal{T}_{n+1}} S_T(r), \tag{6}$$

where, orienting T away from root 0 and writing $p_T(v)$ for the parent of $v \neq 0$,

$$S_T(r) = \sum_{\substack{(Q_0,\dots,Q_n)\in\mathcal{P}^{n+1}\\ r\in\mathrm{skel}(Q_0)}} \left(\prod_{v\neq 0} \mathbf{1}[Q_{p_T(v)} \not\sim Q_v]\right) \prod_{v=0}^{n} u(Q_v). \tag{7}$$

Proof. Insert (1) and (4) into the definition of $H_n^{\#}(Y)$. Apply lemma 3.1, insert the marker (5), use simultaneous permutation invariance to move the marked coordinate to 0, and finally forget the exact-union condition. The normalization is $1/n!$. □

At this point target, root, incompatibility, and holes have been handled without conflation: the target survives through its extracted decay, the root remains pinned, and every tree edge still carries the genuine active-skeleton incompatibility indicator.

4 Fixed-tree leaf summation

For $T \in \mathcal{T}_{n+1}$ let $c_T(v)$ be the number of children of v in the root-0 orientation. Then

$$\sum_{v=0}^{n} c_T(v) = n. \tag{8}$$

Theorem 4.1 (Fixed-tree factorial-moment bound)**.** *Under Assumption 2.2, for every $T \in \mathcal{T}_{n+1}$ and every active root r,*

$$S_T(r) \le M^{2n+1} \prod_{v=0}^{n} c_T(v)!. \tag{9}$$

Proof. Write $\ell_v = \ell(Q_v)$. Parent fibers satisfy the exact cancellation identity

$$\prod_{v\neq 0} \ell_{p_T(v)} = \prod_{v=0}^{n} \ell_v^{c_T(v)}. \tag{10}$$

Therefore the summand in (7) can be rewritten as

$$\left[\prod_{v\neq 0} \frac{\ell_v^{c_T(v)} u(Q_v) \mathbf{1}[Q_{p_T(v)} \not\sim Q_v]}{\ell_{p_T(v)}}\right] \ell_0^{c_T(0)} u(Q_0). \tag{11}$$

Sum the nonroot variables from the leaves toward the root. At a nonroot vertex v, with its parent polymer fixed, the corresponding normalized kernel is bounded by (C_j) with $j = c_T(v)$:

$$\sum_{Q_v \not\sim Q_{p_T(v)}} \frac{\ell(Q_v)^{c_T(v)} u(Q_v)}{\ell(Q_{p_T(v)})} \le c_T(v)! M^{c_T(v)+1}.$$

After all nonroot variables are eliminated, the root sum is bounded by (R_j) with $j = c_T(0)$:

$$\sum_{Q_0 :\, r\in\mathrm{skel}(Q_0)} \ell(Q_0)^{c_T(0)} u(Q_0) \le c_T(0)! M^{c_T(0)+1}.$$

Multiplying the vertex budgets yields

$$S_T(r) \le \left(\prod_v c_T(v)!\right) M^{\sum_v (c_T(v)+1)}.$$

By (8), the exponent is $n + (n+1) = 2n+1$. $\square$

Remark 4.2 (Why the factorial is structural)**.** The factor $c_T(v)!$ is not a coarse tree-count artifact. It is the exact price of the $c_T(v)$-th metric moment generated when the descendants of v have been summed. The remaining combinatorial problem is precisely to evaluate these factorial products over labelled tree shapes.

Combining proposition 3.2 and theorem 4.1 gives

$$e^{\rho\ell(Y)}|H_n^{\#}(Y)| \le \varepsilon^{n+1} M^{2n+1} \left(\frac{1}{n!}\sum_{T\in\mathcal{T}_{n+1}} \prod_v c_T(v)!\right). \tag{12}$$

Everything specific to holes is now confined to M, the target length $\ell(Y)$, and the existence of $r \in \mathrm{skel}(Y)$. The parenthesized expression is pure enumerative combinatorics.

5 The exact rooted-tree identity

Let $\mathcal{O}_n$ denote the set of plane rooted tree shapes with n edges. It is classical that

$$|\mathcal{O}_n| = \mathrm{Cat}_n = \frac{1}{n+1}\binom{2n}{n}; \tag{13}$$

for example, by the first-child/next-sibling correspondence with binary trees or the contour-word bijection with Dyck paths [6, 9].

Theorem 5.1 (Rooted child-factorial identity)**.** *For every $n \ge 0$,*

$$\boxed{\frac{1}{n!}\sum_{T\in\mathcal{T}_{n+1}} \prod_{v=0}^{n} c_T(v)! = \mathrm{Cat}_n\,.} \tag{14}$$

Equivalently,

$$\sum_{T\in\mathcal{T}_{n+1}} \prod_v c_T(v)! = n!\,\mathrm{Cat}_n\,.$$

Proof. For a rooted labelled tree T, independently choose a linear order on the children of each vertex. Since the child set of v has cardinality $c_T(v)$, the number of order assignments is $\prod_v c_T(v)!$. Thus the unnormalized left-hand side counts pairs

$$\mathfrak{D}_n = \{(T, (\prec_v)_{v=0}^n) : T \in \mathcal{T}_{n+1},\ \prec_v \text{ linearly orders the children of } v\}.$$

Orienting T away from 0 and using the chosen child orders turns such a pair into a plane rooted tree whose vertices are labelled by $\{0, 1, \dots, n\}$ and whose root has label 0. Conversely, forgetting the plane embedding recovers T, while the left-to-right child orders recover every $\prec_v$. This is a bijection.

Now forget the nonroot labels. Every shape in $\mathcal{O}_n$ has n nonroot vertex positions, which can be labelled bijectively by $\{1, \dots, n\}$ in exactly $n!$ ways; the root label is fixed. Consequently

$$|\mathfrak{D}_n| = n!|\mathcal{O}_n| = n!\,\mathrm{Cat}_n\,.$$

$\square$

Remark 5.2 (Formalization-friendly factorization)**.** A Lean proof may factor the bijection into three reusable equivalences: a tree together with one permutation of every finite child fiber is a labelled ordered rooted tree; first-child/next-sibling turns an ordered rooted tree with n nonroot vertices into a binary tree with n nodes; fixing the root label leaves $n!$ labelings. Mathlib already supplies Catalan numbers, finite permutation cardinalities, and a Catalan cardinality theorem for binary trees [11].

Corollary 5.3 (Exact improvement over the profile count)**.** *The normalized tree sum obeys*

$$\mathrm{Cat}_n = \frac{1}{n+1}\binom{2n}{n} \le \binom{2n}{n} \le 4^n.$$

Thus a parent-map profile overcount loses an explicit factor $n+1$, and the subsequent geometric bound discards the polynomial decay of the Catalan coefficients.

6 Catalan closure and the square-root majorant

For $M > 0$ and $\varepsilon \ge 0$, define

$$B_M(\varepsilon) := \sum_{n=0}^{\infty} \mathrm{Cat}_n\, M^{2n+1}\varepsilon^{n+1} = M\varepsilon \sum_{n=0}^{\infty} \mathrm{Cat}_n (M^2\varepsilon)^n. \tag{15}$$

Let $C(x) = \sum_{n\ge 0} \mathrm{Cat}_n\, x^n$. The Catalan recurrence gives $C = 1 + xC^2$. For $0 \le x < 1/4$, the branch with $C(0) = 1$ is

$$C(x) = \frac{1-\sqrt{1-4x}}{2x} = \frac{2}{1+\sqrt{1-4x}}, \tag{16}$$

where the second expression also covers $x = 0$ directly.

Theorem 6.1 (Square-root closure)**.** *Let $M > 0$ and $0 \le 4M^2\varepsilon < 1$. Then* (15) *converges and*

$$\boxed{B_M(\varepsilon) = \frac{1-\sqrt{1-4M^2\varepsilon}}{2M}.} \tag{17}$$

Moreover,

$$B_M(\varepsilon) = M\varepsilon + MB_M(\varepsilon)^2, \tag{18}$$

and $B_M(\varepsilon)$ is the least nonnegative solution of this fixed-point equation.

Proof. Set $x = M^2\varepsilon$ in (16) and multiply by $M\varepsilon$, yielding (17). Multiplying $C = 1 + xC^2$ by $M\varepsilon$ gives

$$B_M = M\varepsilon + M(M\varepsilon C(x))^2 = M\varepsilon + MB_M^2.$$

The quadratic $Mb^2 - b + M\varepsilon = 0$ has roots $(1 \pm \sqrt{1-4M^2\varepsilon})/(2M)$. The minus branch is nonnegative, tends to zero with ε, and is no larger than the plus branch. □

Corollary 6.2 (Linear small-activity bound)**.** *If $M > 0$ and $0 \le 4M^2\varepsilon \le 1$, then*

$$0 \le B_M(\varepsilon) = \frac{2M\varepsilon}{1+\sqrt{1-4M^2\varepsilon}} \le 2M\varepsilon. \tag{19}$$

The numerical Catalan series also converges at $4M^2\varepsilon = 1$ and has value $1/(2M)$. We use a strict interior condition below because independent source interfaces often impose strict smallness. Boundary convergence of the Catalan series should not be confused with boundary validity of every model-specific analytic input.

6.1 Explicit comparison with the geometric majorant

The geometric closure obtained from the bound $\mathrm{Cat}_n \le 4^n$ is

$$G_M(\varepsilon) := \sum_{n=0}^{\infty} 4^n M^{2n+1} \varepsilon^{n+1} = \frac{M\varepsilon}{1-4M^2\varepsilon}, \qquad 0 \le 4M^2\varepsilon < 1. \tag{20}$$

Proposition 6.3 (Catalan versus geometric closure)**.** *For $M > 0$ and $0 \le 4M^2\varepsilon < 1$,*

$$B_M(\varepsilon) \le G_M(\varepsilon), \tag{21}$$

with strict inequality for $\varepsilon > 0$. Coefficientwise,

$$\begin{aligned} B_M(\varepsilon) &= M\varepsilon + M^3\varepsilon^2 + 2M^5\varepsilon^3 + 5M^7\varepsilon^4 + \cdots, \\ G_M(\varepsilon) &= M\varepsilon + 4M^3\varepsilon^2 + 16M^5\varepsilon^3 + 64M^7\varepsilon^4 + \cdots. \end{aligned}$$

As $4M^2\varepsilon \uparrow 1$, $B_M(\varepsilon) \to 1/(2M)$ while $G_M(\varepsilon) \to +\infty$.

Proof. The coefficientwise inequality is $\mathrm{Cat}_n \le 4^n$. It is strict for every $\varepsilon > 0$ because $\mathrm{Cat}_1 = 1 < 4$. The limiting statements follow from (17) and (20). □

The Catalan replacement changes neither the fixed-tree moment constant M, nor the target rate, nor the hole geometry. It changes only the final universal shape sum.

7 From $K^\#$ to $H^\#$

Theorem 7.1 (Catalan second-gas bound with prescribed union)**.** *Assume the metric stitching and moment hypotheses, Assumptions 2.1 and 2.2, together with the pointwise first-activity estimate* (1)*. Let Y be a representable nonempty target and choose $r \in \mathrm{skel}(Y)$. If $M > 0$ and $0 \le 4M^2\varepsilon < 1$, then the second Ursell series* (3) *is absolutely convergent and*

$$\boxed{|H^\#(Y)| \le B_M(\varepsilon) e^{-\rho\ell(Y)}.} \tag{22}$$

At fixed order,

$$|H^\#_n(Y)| \le \mathrm{Cat}_n\, M^{2n+1}\varepsilon^{n+1} e^{-\rho\ell(Y)}. \tag{23}$$

Proof. Insert theorem 5.1 into (12) to obtain (23). Summing over $n \ge 0$ gives (22) by the definition of B_M and proves absolute convergence. □

Corollary 7.2 (Appendix-F residual-rate form)**.** *Suppose the decay split in the first activity yields*

$$\rho = \kappa - 3\kappa_0 - 3, \qquad \kappa \ge 3\kappa_0 + 3.$$

Then $\rho \ge 0$ and

$$|H^\#(Y)| \le B_M(\varepsilon) \exp[-(\kappa - 3\kappa_0 - 3)(d_M(Y \bmod \Omega^c) + 1)].$$

In particular, corollary 6.2 gives

$$|H^\#(Y)| \le 2M\varepsilon e^{-\rho\ell(Y)}.$$

The statement preserves the Appendix-F structure: the union is exactly Y, incompatibility is active-skeleton overlap, and decay is measured in the metric with holes contracted. Only after those facts have been used does the proof pass to a root-marked overcount.

8 Rooted profile and one-scale remainder

A pointwise activity estimate becomes useful to a renormalization-group consumer after it has been summed in a local window. Let $a \in \Omega$ and, for $0 \le \lambda \le \rho$, define

$$\mathcal{P}_a^{\lambda}(F) := \sum_{Y:\, a \in \mathrm{skel}(Y)} e^{\lambda \ell(Y)} |F(Y)|. \tag{24}$$

Assumption 8.1 (Rooted hole-metric summability)**.** For some $\kappa_0 > 0$ and $K_0 < \infty$,

$$\sup_{a\in\Omega} \sum_{Y:\, a \in \mathrm{skel}(Y)} e^{-\kappa_0 \ell(Y)} \le K_0. \tag{25}$$

Corollary 8.2 (Rooted Catalan profile)**.** *Under the hypotheses of theorem 7.1 and Assumption 8.1, if $\rho \ge \kappa_0$, then*

$$\sup_{a\in\Omega} \mathcal{P}_a^{\rho-\kappa_0}(H^{\#}) \le K_0 B_M(\varepsilon). \tag{26}$$

Proof. By (22),

$$e^{(\rho-\kappa_0)\ell(Y)} |H^{\#}(Y)| \le B_M(\varepsilon) e^{-\kappa_0 \ell(Y)}.$$

Sum over targets rooted at a and apply (25). □

For the residual rate $\rho = \kappa - 3\kappa_0 - 3$, the sufficient margin $\rho \ge \kappa_0$ is $\kappa \ge 4\kappa_0 + 3$. This is stronger than the nonnegativity condition $\kappa \ge 3\kappa_0 + 3$. The stronger margin is needed only when a summability theorem proved at exponent κ_0 is reused; a direct theorem at the residual exponent avoids this loss.

Let $t, k \in \mathbb{N}$ be scale indices, let $g(k) \ge 0$, and let $a_{t,k}$ be the root selected by the one-scale construction. Suppose the scalar remainder is

$$R_{\mathrm{sc}}(t,k) = \sum_{Y:\, a_{t,k} \in \mathrm{skel}(Y)} \pi(H^{\#}_{t,k}(Y)), \qquad |\pi(z)| \le |z|. \tag{27}$$

Definition 8.3 (Consumer-facing one-scale decay)**.** The predicate `SingleScaleUVDecay` means that

$$|R_{\mathrm{sc}}(t,k)| \le A e^{-c_0 t} g(k)^{\kappa_0} \qquad \text{for every } t,k \in \mathbb{N}. \tag{28}$$

Theorem 8.4 (Catalan profile to a one-scale remainder)**.** *For every t, k, assume the hypotheses of theorem 7.1 with smallness parameter $\varepsilon_{t,k}$ and a common moment constant M. Assume also Assumption 8.1 and*

$$B_M(\varepsilon_{t,k}) \le A e^{-c_0 t} g(k)^{\kappa_0}.$$

If (27) is absolutely summable, then

$$|R_{\mathrm{sc}}(t,k)| \le A K_0 e^{-c_0 t} g(k)^{\kappa_0}. \tag{29}$$

Thus `SingleScaleUVDecay` holds with amplitude AK_0.

Proof. Use the contraction property of π, absolute convergence, and (26) with zero or nonnegative tilt:

$$|R_{\mathrm{sc}}(t,k)| \le \sum_{Y:\, a_{t,k} \in \mathrm{skel}(Y)} |H^{\#}_{t,k}(Y)| \le K_0 B_M(\varepsilon_{t,k}).$$

Then apply (8.4). □

This is the requested pipeline

$$K^{\#} \longrightarrow H^{\#} \longrightarrow \text{rooted profile} \longrightarrow \text{one-scale remainder}.$$

The first arrow contains the Catalan closure; the later arrows are norm and summability transports and therefore remain unchanged when the geometric majorant is replaced.

9 Lean 4 formalization architecture

Lean 4 and mathlib provide the proof-assistant substrate [8, 10]. The proposed formalization is a conservative extension of the base repository state listed in section 10. The base chain already establishes the exact finite fixed-tree estimate and then closes it using 4^n. The Catalan patch is intended to replace only that universal shape theorem and add the square-root analytic consumer.

9.1 Existing verified substrate

Module	Role in the proof
`KP/RootedChildCount`	Child fibers, $\sum_v c_T(v) = n$, factorial products, and explicit child-order assignments.
`KP/RootedLeafSummation`	Current parent-profile and 4^n aggregate tree bounds; this is the replacement point.
`RG/SecondUrsellLeafSummation`	Hard-core moment kernels, exact metric cancellation, fixed-tree bound $M^{2n+1} \prod_v c_T(v)!$, and current geometric leaf closure.
`RG/AppendixFHsharp*`	Fixed-union $H^{\#}$, tree domination, source majorants, profiles, convergence, and residual adapters.
`RG/ClusterWithHolesBridge`	Rooted modified-metric summability and residual-rate aggregation.
`RG/SingleScaleUVDecay`	Absolute activity sum to the scalar one-scale consumer predicate.

Table 1: Existing formal substrate consumed by the Catalan patch.

The fixed-tree result corresponding to theorem 4.1 has the following shape.

```
theorem
  fixedParentKernelSum_le_childFactor_mul_momentPow ... :
  fixedParentKernelSum ... T <=
    (Finset.univ.prod fun v : Fin (n + 1) =>
      ((KP.rootedChildCount T v).factorial : Real)) *
    appendixFSecondUrsellMomentConstant d kappa0 ^ (2 * n + 1)
```

The current aggregate step is the theorem whose name ends in `normalized_le_four_pow`. The exact theorem has the same left-hand side and changes only the right-hand side.

9.2 New reusable API

The pure-combinatorial module is proposed as `YangMills/KP/RootedCatalan.lean`. Its headline statement is:

```
theorem rootedChildCount_factorialTreeSum_normalized_eq_catalan
    (n : Nat) :
```

```
    ((n : Real) + 1) / ((n + 1).factorial : Real) *
      (KP.spanningTrees (top : SimpleGraph (Fin (n + 1))).sum
        (fun T => Finset.univ.prod fun v : Fin (n + 1) =>
          ((KP.rootedChildCount T v).factorial : Real)))
      = (Nat.catalan n : Real)
```

Because $(n+1)/(n+1)! = 1/n!$, this is exactly (14) in the normalization already consumed by the second gas.

The analytic module defines both majorants and proves the fixed-point equation:

```
noncomputable def catalanClosure (M epsilon : Real) : Real :=
  (1 - Real.sqrt (1 - 4 * M^2 * epsilon)) / (2 * M)

noncomputable def geometricClosure (M epsilon : Real) : Real :=
  M * epsilon / (1 - 4 * M^2 * epsilon)

theorem catalanClosure_fixedPoint
    (hM : 0 < M) (hepsilon : 0 <= epsilon)
    (hsmall : 4 * M^2 * epsilon <= 1) :
    catalanClosure M epsilon =
      M * epsilon + M * (catalanClosure M epsilon)^2 := by
  have hrad : 0 <= 1 - 4 * M^2 * epsilon := by nlinarith
  have hs := Real.sq_sqrt hrad
  unfold catalanClosure
  field_simp [ne_of_gt hM]
  nlinarith
```

The formal power-series identity is independently available in mathlib as `PowerSeries.catalanSeries_sq_mul_X_add_one`; the real square-root theorem belongs in the project layer because the polymer consumer is an inequality over $\mathbb{R}$.

The source theorem keeps the established target-decay extraction and leaf sum, replacing its geometric ratio by the exact order bound

```
appendixFHoleHsharpWeightedTreeTerm ... Y n <=
  (Nat.catalan n : Real) * M^(2*n+1) * epsilon^(n+1) *
    appendixFHoleExpWeight HF residualRate Y
```

and its total profile by `catalanClosure M epsilon`. No definition of full support, skeleton, incompatibility, or modified metric changes.

9.3 Proof decomposition in Lean

A reviewable implementation should expose the following intermediate objects rather than hide the bijection in one opaque cardinality calculation.

1. `rootedChildOrderAssignments T`: one finite permutation for each child fiber, whose cardinality is $\prod_v c_T(v)!$.
2. A type of labelled ordered rooted trees with root label 0 and a forgetful map to complete-graph spanning trees.
3. An equivalence between ordered rooted trees and a Catalan-counted tree type, such as binary trees under first-child/next-sibling.
4. A separation of shape and labels, proving that every unlabelled shape has exactly $n!$

root-preserving labelings.

5. A cast theorem from the natural-number cardinality identity to the real normalized identity consumed by the existing second-Ursell code.

This decomposition makes the exact theorem reusable and gives small finite objects for regression testing.

9.4 Oracle discipline

Every new headline theorem is checked with

```
#print axioms YangMills.KP.
    rootedChildCount_factorialTreeSum_normalized_eq_catalan
#print axioms YangMills.RG.appendixFHoleHsharpTerm_le_catalan
#print axioms YangMills.RG.singleScaleUVDecay_of_catalanProfile
```

The expected output is the standard Lean kernel footprint `[propext, Classical.choice, Quot.sound]`, with no `sorryAx` and no project-specific axioms. The finite bijection is classical; use of `Classical.choice` is expected and does not weaken kernel soundness.

10 Reproducibility

A future archival artifact should archive both the paper source and the exact repository commit containing the Catalan patch. The base state inspected for this manuscript is:

Base repository commit	`1d044a353ac2b69ddca732dd851fb0ab4a94d7af`
Lean toolchain	`leanprover/lean4:v4.29.0-rc6`
mathlib commit	`07642720480157414db592fa85b626dafb71355b`
Core target	`YangMillsCore`
Catalan patch status	specified here; not yet an archived Lean commit

From a fresh clone, the base verification commands are:

```
git checkout 1d044a353ac2b69ddca732dd851fb0ab4a94d7af
lake exe cache get
lake build YangMillsCore
lake env lean oracle_check.lean
python scripts/check_consistency.py
```

Once the Catalan patch is implemented, the focused checks are:

```
lake env lean YangMills/KP/RootedCatalan.lean
lake env lean YangMills/RG/AppendixFHsharpCatalanClosure.lean
lake env lean YangMills/RG/AppendixFHsharpCatalanSource.lean
lake build YangMillsCore
lake env lean oracle_check.lean
python scripts/check_consistency.py
```

Any artifact that claims the Catalan patch has been formalized must record the resulting commit rather than a moving branch. It should also include a machine-readable theorem manifest listing the fixed-tree lemma, exact Catalan identity, square-root closure, pointwise $H^{\#}$ theorem, rooted profile, and final `SingleScaleUVDecay` producer.

Remark 10.1 (Status of the code claim)**.** The public base state cited above contains the fixed-tree child-factorial estimate and the geometric closure. The exact Catalan theorem and square-root downstream adapter are the narrowly scoped patch specified by this paper. Publication language such as "formalized in Lean with no `sorry`" should be used for that patch only after the archived commit builds and its axiom report is included in the artifact.

11 What this theorem does not prove

This section is part of the theorem statement, not a disclaimer added after the fact. The formal result has a precise conditional scope.

1. **It does not produce the first activity $K^{\#}$.** The estimate (1), its integrability, support properties, and scale dependence are inputs. In a gauge model they must come from the actual fluctuation integral and localization analysis.
2. **It does not prove `hRpoly` or a model-specific gauge remainder theorem.** The scalar extraction in (27) must be contractive and must represent the physical remainder; that identification is external.
3. **It does not prove the Balaban renormalization-group construction.** Stability, gauge covariance, large-field control, interpolation, random-walk expansions, and iteration through all scales remain separate obligations.
4. **It does not derive the local moment budgets from Yang–Mills dynamics.** This paper proves how (C_j)–(R_j) close the tree sum. Their realization with the desired physical constants must be verified in the concrete model.
5. **It does not take a continuum limit.** No removal of lattice spacing or volume cutoff is performed here.
6. **It does not establish Osterwalder–Schrader or Wightman reconstruction.** Reflection positivity, Euclidean invariance in the limit, regularity, and reconstruction are outside the theorem.
7. **It does not prove the Yang–Mills existence and mass-gap problem.** The endpoint `SingleScaleUVDecay` is one conditional scalar input to a larger lattice argument; it is not a continuum Clay theorem.
8. **It does not certify an unarchived code claim.** The Catalan patch is a formalization claim only after the exact archived commit builds, contains no `sorry`, and passes the stated axiom and consistency checks.

What the theorem *does* prove is correspondingly sharp: once a first activity and the explicit local hole-geometry budgets are supplied, the complete second-Ursell combinatorics – prescribed union, root marking, hard-core edges, exact Catalan tree summation, rooted profile, and scalar one-scale aggregation – closes with the square-root majorant.

12 Discussion

The identity in theorem 5.1 is elementary, but its exact use has three advantages in a formal cluster expansion. First, it exposes the real combinatorial object generated by the leaf proof: a plane ordering at each parent, not an arbitrary parent-map profile. Second, it separates constants arising from geometry from the universal tree-shape coefficients. Third, it provides a canonical nonlinear closure, $B = M\varepsilon + MB^2$, that can be reused in other polymer systems.

The improvement is especially visible near the nominal geometric threshold. The geometric majorant has a pole, whereas the Catalan closure remains finite. This does not automatically enlarge a physical convergence domain because the source activity and other analytic interfaces may impose stricter inequalities. It does remove an artificial divergence introduced solely by replacing Cat_n with 4^n.

Formalization work in constructive quantum field theory is now substantial [3]. The contribution here is complementary: a compact, source-shaped combinatorial kernel is connected to an existing renormalization-group API. The appropriate claim is neither that Catalan numbers are new nor that Yang–Mills is solved, but that a delicate cluster-expansion pipeline has been stated at the correct abstraction boundary and can be certified end to end once its model-specific inputs are supplied.

A A recursive proof of the Catalan identity

For completeness, define

$$a_n = \frac{1}{n!} \sum_{T \in \mathcal{T}_{n+1}} \prod_v c_T(v)!.$$

By the child-order interpretation, a_n is the number of plane rooted tree shapes with n edges. Decompose a nontrivial plane rooted tree using the standard first-child/next-sibling binary encoding. The root decomposition gives

$$a_0 = 1, \qquad a_{n+1} = \sum_{i=0}^{n} a_i a_{n-i}.$$

These are the Catalan recurrence and initial value, so $a_n = \mathrm{Cat}_n$. The generating function $A(x) = \sum_{n \geq 0} a_n x^n$ satisfies $A = 1 + xA^2$, exactly the equation used in theorem 6.1. This second proof is useful in Lean when a recursive ordered tree datatype is more convenient than an explicit global bijection.

B First coefficients and executable sanity checks

For small n, the exact weighted sums are

n	$\sum_T \prod_v c_T(v)!$	$n!\,\mathrm{Cat}_n$	normalized value
0	1	1	1
1	1	1	1
2	4	4	2
3	30	30	5
4	336	336	14

For $n = 2$, the rooted star contributes 2 and the two rooted paths contribute 1 each, giving $4 = 2!\,\mathrm{Cat}_2$. These values are suitable as finite regression tests before the general bijection theorem is invoked.

C Minimal theorem manifest for an archival artifact

A reviewable release should expose at least the following declarations, with names adapted only if the repository namespace changes:

1. `rootedChildCount_factorialTreeSum_normalized_eq_catalan`;
2. `appendixFHoleHsharpWeightedTreeMarkedRootSum_le_catalan`;
3. `appendixFHoleHsharpWeightedTreeTerm_le_catalan`;
4. `catalanClosure_eq_tsum`;
5. `catalanClosure_fixedPoint`;
6. `catalanClosure_le_geometricClosure`;
7. `appendixFHoleHsharp_norm_le_catalanClosure`;
8. `rootedAppendixFHsharpProfile_of_catalanClosure`;
9. `singleScaleUVDecay_of_catalanProfile`.

For each declaration, `oracle_check.lean` should print only the kernel footprint `[propext, Classical.choice, Quot.sound]`.

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